Problem 1)

e) The results demonstrate that meaningful model compression is achievable through strategic combination of techniques rather than relying on any single method.

Among the approaches evaluated, the knowledge distillation student model produced the smallest footprint by targeting the root cause of model size: parameter count. Pruning, while effective at zeroing weights, does not reduce file size in standard TFLite deployments because sparse weights are still stored at full precision. Quantization alone compresses existing weights but cannot eliminate them. Architecture reduction through distillation addresses both simultaneously.

The most effective strategy was stacking knowledge distillation with INT8 quantization. Distillation reduced the parameter count by training a smaller 16→8→3 network guided by the teacher's soft probability outputs, which carry richer gradient signal than hard one-hot labels and allow the compact student to generalize better than if trained from scratch. INT8 quantization then compressed each remaining weight from 32-bit to 8-bit, yielding an additional 4× reduction.

Critically, the Wine dataset is well-suited to aggressive compression. Its 13 features are highly discriminative across three linearly separable classes, meaning even shallow networks capture sufficient decision boundaries. Datasets with greater class overlap or higher-dimensional inputs would show steeper accuracy degradation under equivalent compression, requiring a more conservative trade-off between sparsity, quantization, and architecture depth.

Problem 2)

1.

Edge Impulse's free tier offers three categories of learning blocks. Supervised classification includes neural network classifiers for image, audio, motion, and tabular data, as well as transfer learning blocks like MobileNet. Supervised regression trains a network to predict continuous outputs from sensor data. Anomaly detection (unsupervised) uses K-Means clustering to flag out-of-distribution samples, with the GMM anomaly block as an alternative. Self-supervised learning is not available on the free tier. Examples: a keyword-spotting classifier (supervised), a motion regression model predicting angle, and a K-Means anomaly detector for vibration monitoring.

2.

The free-tier Classifier section supports: Dense (Fully Connected) — learns global feature mappings between all inputs and outputs, used in every network; Dropout — randomly deactivates neurons during training to reduce overfitting; Flatten — reshapes multidimensional tensors into 1D vectors before dense layers; Conv1D — extracts local temporal patterns from sequential sensor data; Conv2D — extracts spatial features from images; MaxPooling1D/2D — downsamples feature maps by retaining maximum activations, reducing dimensionality; Reshape — reorganizes tensor dimensions without changing data. These layers can be stacked in the visual editor to build custom architectures.

3.

Under the Deployment section, Edge Impulse free tier offers: Quantization-aware training producing an optimized INT8 model; float32 uncompressed baseline export; and EON Tuner, which automatically searches architectures and compression settings for a target RAM/ROM budget. Models can be exported as Arduino libraries, C++ source, or TFLite flatbuffers. The INT8 option mirrors post-training quantization, reducing weight storage by 4× and accelerating inference on microcontrollers with fixed-point hardware units like the Cortex-M4F and Cortex-M33.

4.

Free-tier Edge Impulse supports synthetic data generation for audio (via text-to-speech and noise synthesis), images (via object placement on random backgrounds and augmentation pipelines), and motion/IMU (via physics-based movement simulation). Synthetic data is critical in embedded ML because real-world collection is expensive, time-consuming, and often class-imbalanced — exactly the challenge encountered in small datasets like sword strike classification. Synthetic samples improve minority-class representation, reduce overfitting, and enable training without physical hardware access, accelerating prototyping while maintaining reasonable generalization to real sensor distributions.